App Dev Project Report

1. Student Details

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About Me: I am a student in the IIT Madras BS Degree program with a interest in full-stack web development and database management. I enjoy building secure, role-based applications that solve real-world administrative and logistical challenges. Also perusing offline B.Tech in Computer Science Engineering at Nirma University Ahmedabad.

2. Project Details

- **Project Title:** PlaceHub - Campus Placement Portal Application
- **Problem Statement:** Institutes currently rely on manual processes, spreadsheets, and emails to manage campus recruitment, making it difficult to track student applications, manage company approvals, and prevent duplicate entries.
- **Approach:** I built a role-based, centralized web application using Flask. The system strictly separates functionalities for three user types: Admins (Institute), Companies, and Students. It features a secure backend validation system, dynamic interactive dashboards, and real-time application tracking to streamline the entire recruitment lifecycle.

3. AI/LLM Declaration

- I used Google Gemini to assist in writing SQLAlchemy model definitions, structuring the Flask routing logic, and designing the CSS for the modal overlays and dashboard layouts.
- The extent of AI/LLM usage is approximately 20–30%, primarily used for code scaffolding, CSS variable structuring, and debugging database relationship errors.
- All final business logic, database integration, security validations (like the dynamic eligibility engine), and extensive testing were implemented and verified manually.

4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Core backend web framework and routing
SQLAlchemy	Object Relational Mapper (ORM) for database interactions
SQLite	Lightweight local database for storing application data
Jinja2	Template engine for rendering dynamic HTML pages
Custom CSS / HTML5	Frontend styling, responsive design, and CSS variables
JavaScript (Vanilla)	Dynamic UI elements (Search filtering, Password Strength Meter, Modals)
Werkzeug.security	Secure password hashing and verification

5. Database Schema / ER Diagram

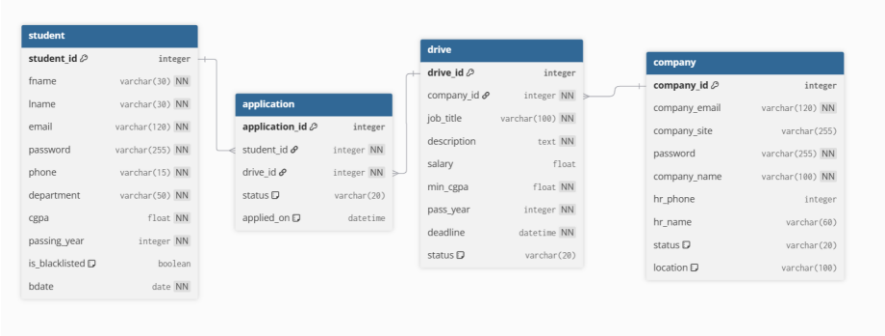
Tables:

- **Student:** Stores student profiles (student_id, fname, lname, email, password, cgpa, passing_year, is_blacklisted).
- **Company:** Stores recruiter profiles (company_id, company_name, company_email, password, hr_name, status, location).
- **Drive:** Logs job postings created by companies (drive_id, company_id, job_title, min_cgpa, pass_year, status).

- **Application:** Tracks student applications to drives (application_id, student_id, drive_id, applied_on, status).

Relationships:

- One-to-Many → **Company** → **Drive** (A company can host multiple drives)
- One-to-Many → **Student** → **Application** (A student can apply to multiple drives)
- One-to-Many → **Drive** → **Application** (A drive can have multiple student applications)



6. Key Application Endpoints

Endpoint	Method	Description
/check	POST	Authenticates user credentials and initiates role-based sessions
/registerStudent	POST	Validates and registers a new student account
/createDrive	POST	Allows an approved company to create a new placement drive
/apply_drive/<drive_id>	POST	Creates an application record for a student
/admin/approve_company/<id>	POST	Admin action to approve a pending company registration

7. Architecture and Features

Architecture Overview:

- app.py – Main application entry point, routing, and controller logic.
- models.py – Database schemas defined using SQLAlchemy.
- /templates – Jinja2 HTML templates separated by user roles (e.g., admin_dashboard.html, student_drives.html).
- /static – Storage for dynamic media including profile photos, company logos, and student resumes.

Implemented Core Features:

- Strict Role-Based Access Control (Admin, Company, Student).
- Dynamic Placement Drive creation and management.
- Real-time Student Application tracking and status updates.
- Admin dashboard with statistical counters and live search filtering.

Additional / Advanced Features:

- **Intelligent Eligibility Engine:** Automatically hides drives from students who do not meet the minimum CGPA or passing year requirements.
- **Bi-Directional Blacklisting:** Instantly revokes visibility of a blacklisted company's drives for students, and hides blacklisted students from companies.
- **Dynamic Password Strength Meter:** Real-time visual feedback enforcing complex passwords during registration.

8. Video Presentation

- **Drive Link:** https://drive.google.com/file/d/1ThfIohCqYxyrH0bwo_IvmfLgkzkHtsNv/view?usp=drive_link